1.7 Inverse Trigonometric Functions

Name _____ Period ____ Date _____

Pre-Calculus 1.7 (30-minute plan) · Student sheet

I can…

- evaluate inverse trigonometric functions exactly and with a calculator.
- use the inverse properties of trigonometric functions.
- evaluate compositions of trigonometric and inverse trigonometric functions.

Vocabulary · fill in the missing word

Inverse sine function — y = _____ x means sin y = x, with $-\pi/2 \le y \le \pi/2$

Inverse cosine function — y = _____ x means cos y = x, with $0 \le y \le \pi$

Inverse tangent function — y = _____ x means tan y = x, with $-\pi/2 < y < \pi/2$

Part 1 · Notes and examples (fill in each note, then work the example)

arcsin x gives the angle in $[-\pi/2, \pi/2]$ whose sine is x. **arccos x** gives the angle in _____.

Example 1

Evaluate.

a. $\arcsin\left(-\frac{1}{2}\right)$

b. $\arccos(-1)$

c. $\arctan 1$

d. $\arcsin 2$

Inverse properties: $\sin(\arcsin x) = x$ for $-1 \le x \le 1$; $\arcsin(\sin y) = y$ only for y in _____.

Example 2

Evaluate, if possible.

a. $\tan(\arctan(-5))$

b. $\arcsin\left(\sin \frac{5\pi}{3}\right)$

c. $\cos(\cos^{-1} \pi)$

Compositions: draw a right triangle for the inside angle, then use _____.

Example 3

Find the exact value.

a. $\tan(\arccos \frac{2}{3})$

b. $\cos(\arcsin(-\frac{3}{5}))$

Part 2 · Partner practice

Partner practice

Find $\sin(\arctan \frac{3}{4})$.